Transformers, Parallel Computation, and Logarithmic Depth

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Spotlight Paper

Motivation

Motivating question: How can transformers be understood as models of parallel computation (e.g. MPC)?

1. Representational powers of depth?
2. Multi-step reasoning capabilities?
3. Comparisons with other models (state-space models, sub-quadratic attention transformers, GNNs)?

Transformer Architecture

- *Self-attention unit:*

$$f(X) = \mathrm{softmax}(XQK^TX^T)XV$$

for input $X \in \mathbb{R}^{N\times d}$ and model parameters $Q, K, V \in \mathbb{R}^{d\times m}$.

- *Multi-headed attention:* $L(X) = X + \sum_{h=1}^{H} f_h(X)$.
- *Arbitrary element-wise encoding (MLP + positional encoding):* $\varphi(X) = (\varphi(x_1, 1), \ldots, \varphi(x_N, N))$.
- *Full transformer:* $T(X) = (\varphi_D \circ L_D \circ \cdots \circ L_1 \circ \varphi_0)(X)$.
- Key assumptions: $m, H, L \ll N$; fixed precision; arbitrary φ.

Massively Parallel Computation (MPC)

Theoretical model of MapReduce by Karloff, et al '10.

1. N-bit input divided among q machines with local memory s.
2. For round $r = 1, \ldots R$:
   - 2.1 Each machine performs some computation on its input state.
   - 2.2 Each machine simultaneously sends and receives messages of size $\leq s$.

(γ, δ)-**MPC protocols** have local memory $s = O(N^\delta)$ and global memory $qs = O(N^{1+\gamma})$.

Conjecture (1-cycle vs 2-cycle; Ghaffari, et al '19)

Any (γ, δ)-MPC protocol with $\delta \in (0, 1)$ that distinguishes one length-N cycle from two length-$\frac{N}{2}$ cycles uses $R = \Omega(\log N)$ rounds.

k-Hop Induction Heads

Induction Heads: Toy task to complete bigram via previous occurrence: $\mathrm{IH}(\mathrm{baebcabebdea}) = \mathrm{b}$.

- Occurs naturally in trained transformers; useful primitive for mechanistic interpretability research [Anthropic blog post].
- Two-step logical reasoning task with natural two-layer transformer construction [Bietti, et al '23].
- Negative result from communication complexity: Single-layer transformers cannot solve IH.

k-Hop Induction Heads: Iterated variant of IH to capture k-step reasoning.

$\mathrm{hop}_k(\ldots a_k a_{k+1} \ldots a_{k-1} a_k \ldots a_1 a_2 \ldots a_1) = a_{k+1}$

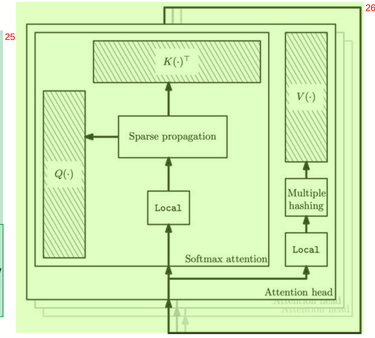
baebcabebdea.

Theory Result #1: Near-equivalence of Transformers & MPC

Theorem (Transformers simulate MPC)

Any R-round (γ, δ)-MPC protocol (with $\delta > \gamma$) can be simulated by a transformer of depth $L = O(R)$ and width $m = \tilde{O}(N^{4\delta})$.

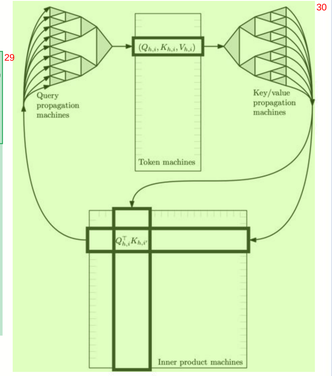
- **Implications:** $O(\log(N))$-depth solutions to graph connectivity, min-spanning forest, cycle check, formula evaluation, etc.
- **Proof Idea:** Simulate each machine's computations in MLPs φ; simulate communication with self-attention units.
- **Subsequent improvement:** Width dependence improved to $m = O(N^{\delta+\varepsilon})$ for any fixed ε by [SFH+24].



Theorem (MPC simulates Transformers)

Any transformer of depth L and width m can be simulated by an $(1 + \delta, \delta)$-MPC protocol with $R = O(L)$ rounds, $s = O(m)$ local memory, and $q = O(N^2)$ machines.

- **Implications:** Under 1-cycle vs 2-cycle conjecture, $L = \Omega(\log N)$ depth is necessary to solve graph connectivity.
- **Proof Idea:** $O(N)$ "token machines" compute $Q/K/V$ embeddings; $O(N^2)$ "inner product machines" compute all Q/K inner products.
- **Limitation:** Not exactly bidirectional relationship between theorems due to $q = O(N^2)$ machine requirement.



Theory Result #2: Efficient Multi-Step Reasoning Representations

Construction: Transformers of depth $L = \lfloor \log_2 k \rfloor + 2$ and width $m = O(1)$ can solve k-IH.

- First two layers solve induction heads for each input.
- Subsequent layers employ "pointer doubling" to chain together solutions.

Optimality: All transformers of width $m = O(N^{1-\varepsilon})$ that solve k-IH have depth $L = \Omega(\log k)$ (conditional on 1-cycle vs 2-cycle conjecture).

- Reduction to MPC simulation of transformers.

Theory Result #3: Sub-optimality of Other Architectures

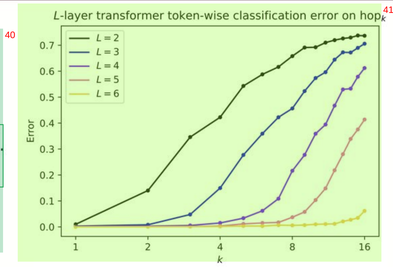
Graph Neural Networks: Solving graph connectivity requires $L\sqrt{m} = \tilde{\Omega}(N^{1/4})$ [CONGEST reduction; Loukas '19].

State-Space Models: RNN, LSTM, Mamba solutions to k-IH requires depth $L \geq k$ or width $m = \Omega(N/k^6)$ [Pointer chasing reduction; Assadi, N '21].

Sub-Quadratic Attention Transformers: Every kernel-based (e.g. Performer) or masking-based (e.g. Longformer) transformer variant that solves k-IH has either depth $L \geq k$ or near-quadratic attention computations [Pointer chasing].

Empirical Result #1: Log-depth k-Hop Induction Heads

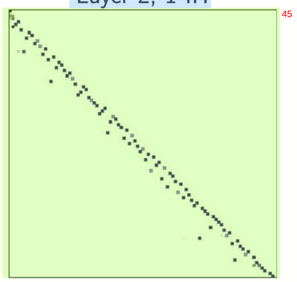
- **Task:** Multi-task k-IH for $k = 0, 1, \ldots, 16$; sequence length $N = 100$.
- **Models:** GPT-2 transformers with embed. dim. $m = 128$, depth $L = 2, \ldots, 6$.
- **Result:** Low error when $L \leq \lfloor \log_2 k \rfloor$, as predicted by theory.



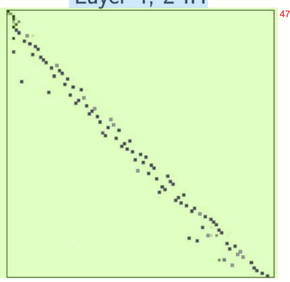
Empirical Result #2: Interpretability of Learned Models

16-IH attention matrices $\mathrm{softmax}(XQK^TX^T)$, with colored partial solutions

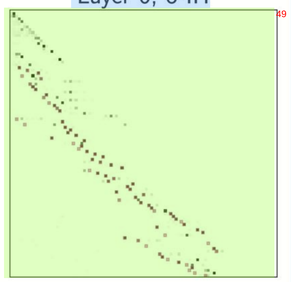
Layer 2, 1-IH



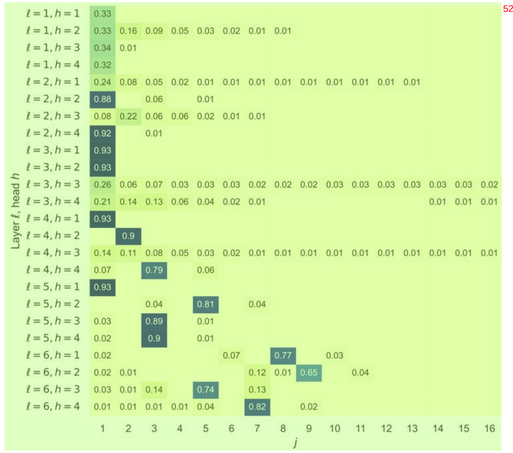
Layer 4, 2-IH



Layer 6, 8-IH



⟨Emp. attn. matrix, theoretical j-IH map⟩



- **Key question:** Do the trained transformers solve k-IH with the same "pointer doubling" approach of the theoretical construction?
- **Yes.** Close correspondence between self-attention matrices and partial solutions to k-IH.
- At least one head in the ℓth attention layer implements $2^{\ell-2}$-IH.

Paper Links

Prior communication compl. + attention paper. [SHT23], NeurIPS '23.

This paper. [SHT24].

Subsequent application to graph algorithms. [SFH+24].